

HOYDALSMO II, Norway

WMO: 014470

Lat: 59.4969N

Lon: 8.1992E

Elev: 562

StdP: 94.75

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.4	-18.9	-23.7	0.5	-21.3	-21.0	0.6	-18.7	6.7	1.7	5.9	0.6	1.1	310	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	24.2	14.5	22.3	13.9	20.2	13.1	16.0	20.7	15.1	19.3	14.1	18.2	2.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.5	11.0	16.9	13.5	10.3	16.1	12.4	9.6	15.2	46.6	20.7	44.0	19.4	41.2	18.1	19.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.7	4.9	4.2	-25.7	26.4	2.5	1.6	-27.5	27.6	-29.0	28.5	-30.4	29.4	-32.3	30.6		
(5)			-25.9	17.3	2.1	0.9	-27.4	18.0	-28.7	18.5	-29.8	19.0	-31.4	19.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-6.8	-5.5	-2.1	2.4	7.1	11.1	13.4	11.6	8.4	3.4	-1.7	-5.7	(6)
(7)		DBStd	8.06	5.13	5.19	4.71	3.36	3.96	2.86	3.01	2.24	2.72	3.17	4.38	5.81	(7)
(8)		HDD10.0	2799	520	433	375	227	108	19	4	9	61	207	351	486	(8)
(9)		HDD18.3	5589	778	666	633	477	347	217	154	210	297	464	601	744	(9)
(10)		CDD10.0	255	0	0	0	0	20	53	110	57	14	1	0	0	(10)
(11)		CDD18.3	2	0	0	0	0	0	0	2	0	0	0	0	0	(11)
(12)		CDH23.3	84	0	0	0	0	4	19	60	1	0	0	0	0	(12)
(13)		CDH26.7	4	0	0	0	0	0	1	3	0	0	0	0	0	(13)
(14)	Wind	WSAvg	1.7	1.6	1.7	1.9	1.8	1.9	2.0	1.7	1.6	1.7	1.6	1.4	1.5	(14)
(15)	Precipitation	PrecAvg	949	87	51	49	53	66	78	97	107	92	105	91	74	(15)
(16)		PrecMax	1421	176	134	97	127	169	156	236	183	181	274	306	156	(16)
(17)		PrecMin	711	10	9	4	16	5	9	8	7	42	21	21	7	(17)
(18)		PrecStd	151	41	32	26	31	38	39	50	47	38	58	57	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	7.6	12.8	17.1	23.8	25.7	27.1	22.7	20.5	14.6	9.1	7.5	(19)
(20)			MCWB	3.1	3.8	6.0	9.1	13.5	13.7	15.4	14.9	14.0	11.3	7.1	5.1	(20)
(21)		2%	DB	4.0	5.5	9.6	14.6	21.1	23.1	25.1	20.7	17.4	12.3	7.2	5.6	(21)
(22)			MCWB	2.1	2.7	5.1	7.9	12.2	12.8	15.2	14.4	12.0	8.7	5.7	3.7	(22)
(23)		5%	DB	2.1	3.7	6.8	12.1	18.3	20.9	23.2	18.7	15.5	10.4	5.6	3.8	(23)
(24)			MCWB	0.8	1.4	3.5	6.3	10.5	12.2	14.8	13.0	11.2	7.6	4.3	2.3	(24)
(25)		10%	DB	0.7	1.9	5.0	9.3	15.7	18.5	21.0	17.0	13.7	8.8	4.0	2.0	(25)
(26)			MCWB	-0.3	0.2	1.7	4.9	9.5	11.2	13.9	12.4	10.3	6.6	2.9	0.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	4.7	7.1	9.9	14.7	15.1	17.8	16.2	14.8	12.6	7.9	5.4	(27)
(28)			MCDWB	5.3	7.1	12.0	16.3	21.4	20.6	22.7	20.0	17.0	14.7	8.9	7.3	(28)
(29)		2%	WB	2.3	2.9	5.1	8.4	13.2	14.0	16.6	15.3	13.1	9.7	5.9	3.9	(29)
(30)			MCDWB	3.7	5.0	8.8	13.6	18.8	19.9	22.0	18.5	15.6	11.0	7.0	5.4	(30)
(31)		5%	WB	1.0	1.7	3.7	6.8	11.6	13.2	15.7	14.3	12.1	8.4	4.5	2.3	(31)
(32)			MCDWB	1.9	3.3	6.8	11.0	16.5	18.6	20.5	17.5	14.8	9.9	5.6	3.6	(32)
(33)		10%	WB	-0.1	0.5	2.2	5.4	10.2	12.1	14.8	13.3	11.0	6.9	3.0	1.0	(33)
(34)			MCDWB	0.7	1.7	4.5	9.2	14.3	17.0	19.1	16.1	13.2	8.6	3.9	1.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	9.7	11.0	10.8	11.2	11.1	10.4	9.8	9.0	8.2	7.1	7.7	(35)
(36)			MCDBR	8.6	11.0	12.9	15.6	15.5	15.9	15.0	12.9	12.6	10.8	8.5	8.2	(36)
(37)			MCWBR	7.7	8.7	8.4	9.2	7.6	7.2	6.5	6.9	7.6	7.3	6.8	6.8	(37)
(38)		5% WB	MCDWR	8.4	9.6	12.2	14.4	13.0	12.8	11.6	10.3	10.0	8.6	7.4	7.9	(38)
(39)			MCWBR	7.4	7.6	8.1	8.9	6.9	6.6	5.9	6.3	6.7	6.6	6.4	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.230	0.267	0.297	0.336	0.338	0.329	0.347	0.341	0.313	0.297	0.249	0.205	(40)	
(41)		taud	2.194	2.278	2.268	2.288	2.432	2.493	2.453	2.460	2.537	2.498	2.287	2.102	(41)	
(42)		Ebn at Noon	596	760	841	866	892	906	878	856	830	729	568	478	(42)	
(43)		Edh at Noon	41	66	92	108	101	97	99	91	72	55	38	30	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.34	0.99	2.45	4.03	4.79	5.41	4.75	3.75	2.44	1.16	0.43	0.22	(44)	
(45)		RadStd	0.07	0.22	0.30	0.64	0.67	0.51	0.63	0.29	0.32	0.22	0.11	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page